Proyecto del Plano Inclinado

Determinación teórica y experimental del tiempo de descenso de un bloque con fricción cinética. Reporte generado el 3/10/2026.

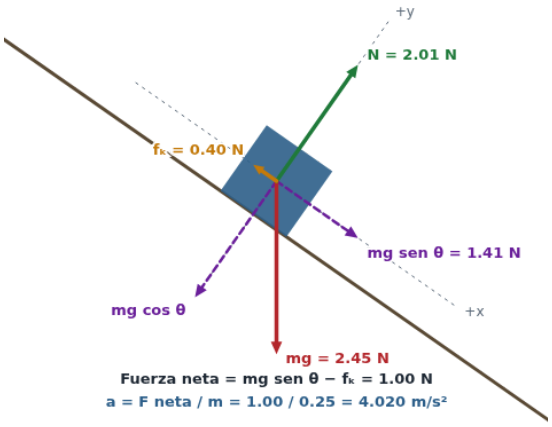
1. Parámetros

θ	L	μ_k	g	v_0	m
35°	1.00 m	0.20	9.81 m/s²	0 m/s	0.25 kg

2. Análisis de fuerzas y cálculo teórico

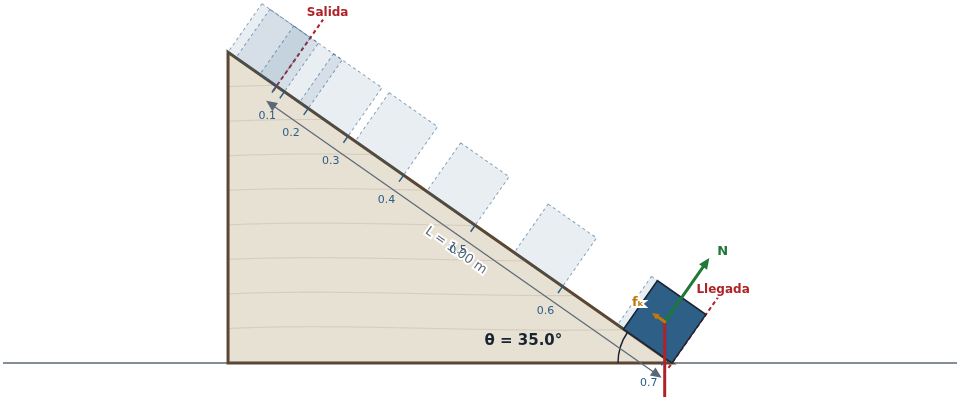
N = mg cos θ; ma = mg sen θ − μ_k mg cos θ ⟹ a = g(sen θ − μ_k cos θ) = 9.81(0.5736 − 0.1638) = **4.020 m/s²**

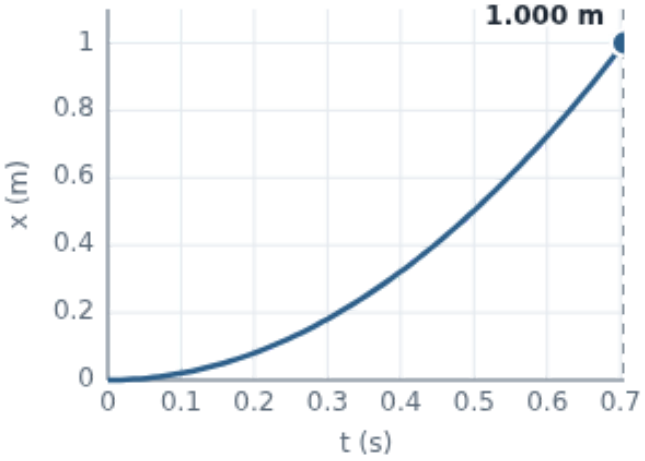
t = √(2L/a) = √(2 × 1.00 / 4.020) = **0.705 s**; v = a·t = **2.84 m/s**

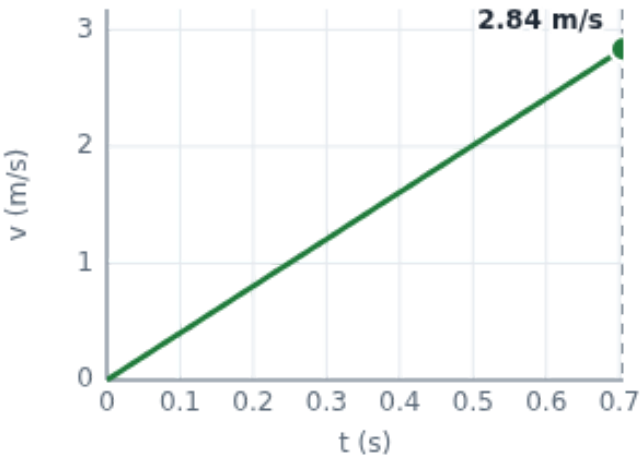


¡Llegó! t = 0.705 s, v = 2.84 m/s

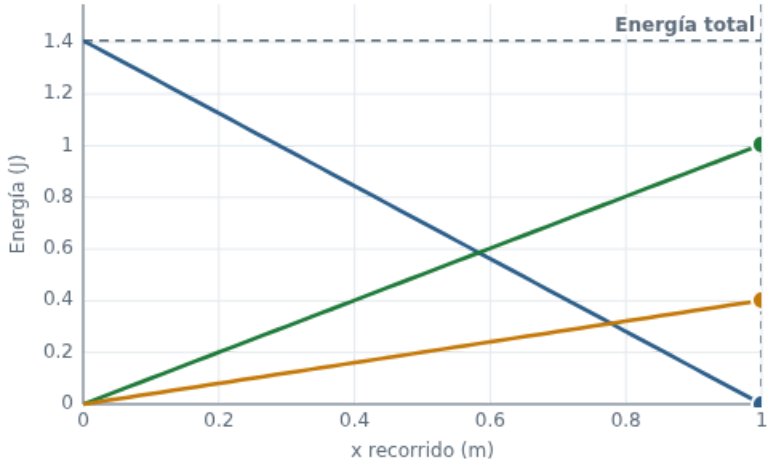
Cronómetro
0.705 s




1.000 m
x (m)
t (s)
0
0.2
0.4
0.6
0.8
1
0.1
0.3
0.5
0.7

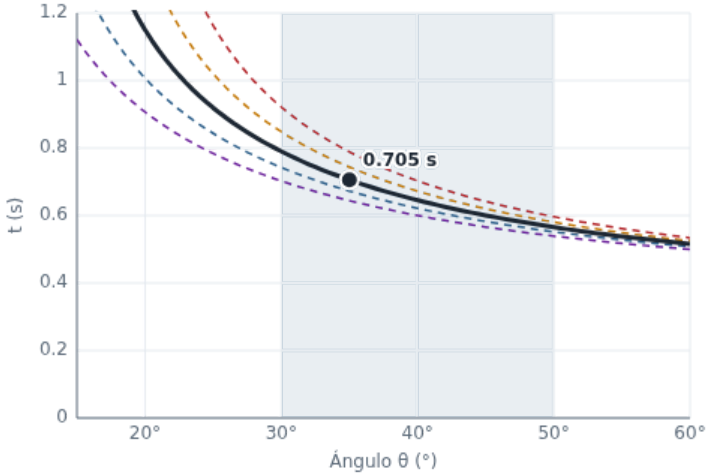

2.84 m/s
v (m/s)
t (s)
0
1
2
3
0.1
0.2
0.3
0.4
0.5
0.6
0.7

3. Energía



4. Sensibilidad (L = 1.00 m)

θ	μ_k = 0.15	μ_k = 0.20	μ_k = 0.25	μ_k = 0.30
30°	0.742 s	0.790 s	0.848 s	0.921 s
35°	0.673 s	0.705 s	0.744 s	0.789 s
40°	0.621 s	0.645 s	0.672 s	0.703 s
45°	0.582 s	0.600 s	0.620 s	0.642 s
50°	0.552 s	0.566 s	0.580 s	0.596 s



5. Registro experimental

Ensayo	Tiempo medido (s)
1	0.740
2	0.690
3	0.780
4	0.710
5	0.760
Promedio	**0.736**

Error (%) = $|t_{teórico} - t_{experimental}| / t_{experimental} \times 100 = |0.705 - 0.736| / 0.736 \times 100 =$ **4.2 %**

Desviación estándar: 0.036 s. Nota según la escala: **100**. μ_k experimental: 0.241.

